



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-054A SolarFleet Lakehouse Gate

Protected independent phase Gate A.

Artifact	C3-054A
Status	2026 - LEARNER EDITION
Phase	C3-04 Storage, File Formats & Lakehouse 13 lessons 34-46 focused hours
Prerequisite	C3-03 Gate PASS; Bridge warehousing/lakehouse mental models assumed, implementation depth taught here.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: mental picture -> tiny example -> guided real file/table workflow -> break/fix -> independent changed task -> validation -> production reasoning -> evidence. A file extension or successful write is never sufficient proof of correct table state.



SolarFleet scenario

SolarFleet receives a base sensor-reading dataset plus a correction batch. The correction updates one `reading_id`, inserts one new row and adds `calibration_batch`. Build and defend a reliable analytical-table path.

Begin only after ColdChain Mini-Lab PASS and all 13 lesson mastery records are complete.

Independent rules

- Use supplied case ZIP; never edit fixtures to manufacture the expected control.
- Lesson/review/video answers are closed during the attempt; only permitted syntax references may be used.
- Declare business key/current-state grain and physical/logical table contract before implementation.
- Save real Parquet/Delta version/schema/MERGE/history evidence; do not substitute planned output for executed output.
- No destructive vacuum is required for the Gate.

Required deliverables

- Choose and defend analytical file format plus partition strategy from workload/cardinality.
- Build current-state transactional table from base data.
- Apply correction with source uniqueness + grain-safe MERGE/upsert.
- Handle additive schema field deliberately and state incompatible-change policy.
- Prove key uniqueness plus source-to-target row/metric reconciliation.
- Read a prior committed version and explain exact change.
- Measure file health; define compaction trigger + post-rewrite parity.
- Define retention window preserving required recovery/time travel.
- Compare Delta and Iceberg conditionally across at least three decision factors.
- Provide table contract + maintenance/recovery runbook.

Competency Gate - every required competency must pass

Competency	PASS evidence
Physical layout / format / partition reasoning	Observed, validated, reproducible and explainable on this fresh case.
Real Parquet metadata and file-health evidence	Observed, validated, reproducible and explainable on this fresh case.
Transactional Delta version/log evidence	Observed, validated, reproducible and explainable on this fresh case.
Controlled schema evolution and contract change	Observed, validated, reproducible and explainable on this fresh case.
Grain-safe MERGE/upsert with reconciliation	Observed, validated, reproducible and explainable on this fresh case.
Historical version read and retention reasoning	Observed, validated, reproducible and explainable on this fresh case.
Layer/catalog/open-table-format architecture and maintenance handoff	Observed, validated, reproducible and explainable on this fresh case.

PASS only if every required competency passes AND zero critical failures remain. Every required competency stands on its own; one failure blocks PASS.



Evidence packet

Evidence	Your saved proof/path/note
Format/layout decision	
Parquet metadata/health evidence	
Delta version/log evidence	
Schema-change record	
MERGE/upsert metrics + key check	
Row/metric reconciliation	
Historical version comparison	
Layer/catalog/table contract	
Compaction + retention plan	
Open-table-format decision	
Recovery/maintenance runbook	